Is Time Ticking?[1]

Online Appendix

Contents

A Proofs **2**

A.1 Proof of Proposition 2.1 (Qualitative Divergence) 2

A.2 Proof of Proposition 4.4 (KT Re-emergence) 4

A.3 Proof of Proposition 4.5 (Deterministic KT) 8

A.4 Proof of Proposition 4.6 (Convergence and Interpolation) 11

A.5 Proof of Proposition 4.7 (Precautionary Under-Investment) 15

B Computational Details **18**

B.1 Grid Specification 18

B.2 Phase 1: HJB Equation 19

B.3 Phase 2: Commitment Optimization 20

B.4 Kolmogorov Forward Equation 21

B.5 General Equilibrium 22

B.6 Convergence Criteria and Verification 23

C Extensions **23**

C.1 The Delay Differential Equation Formulation 24

C.2 Stochastic Build Time 25

C.3 Partial Reversibility 25

C.4 Multiple Commitment Types 26

C.5 Connection to the Krusell-Smith Forecasting Problem 26

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A Proofs

A.1 Proof of Proposition 2.1 (Qualitative Divergence)

Proof. We solve the three-date model of Section 2 in full detail for both Model F and Model C.

Model F. The agent solves

$$\max_{c_0,c_1,c_2,i,s_0,s_1\geq 0} \ln c_0 + \beta \ln c_1 + \beta^2 \ln c_2$$

subject to the budget constraints in Section 2. Substituting the budget constraints, the problem is equivalent to choosing $i \geq 0$ and the split of lifetime resources $R(i) \equiv W + y + f(i) - i$ across three dates, subject to $s_0 \geq 0$ and $s_1 \geq 0$.

The lifetime budget constraint is $c_0 + c_1 + c_2 = R(i)$. With log utility and gross return 1 on storage, the Euler equations give $c_1/c_0 = \beta$ and $c_2/c_1 = \beta$, so

$$c_0 = \frac{R(i)}{1+\beta+\beta^2}, \quad c_1 = \frac{\beta\, R(i)}{1+\beta+\beta^2}, \quad c_2 = \frac{\beta^2\, R(i)}{1+\beta+\beta^2}.$$

The optimal investment maximizes $R(i) = W + y + f(i) - i$, giving the first-order condition

$$f'(i^F) = 1 \quad \Longrightarrow \quad \alpha A (i^F)^{\alpha-1} = 1 \quad \Longrightarrow \quad i^F = (\alpha A)^{1/(1-\alpha)}.$$

This expression depends on A and α but not on W or y.

We verify that the savings constraints are slack when W is large. We have

$$s_0 = W - c_0^F - i^F = W - \frac{R(i^F)}{1+\beta+\beta^2} - i^F,$$

which is positive for W sufficiently large. Similarly,

$$s_1 = s_0 + y + f(i^F) - c_1^F,$$

which is also positive for W large. The savings constraints do not bind, and the unconstrained allocation is achieved.

Model C: Regime 0 (Unconstrained). If neither $s_0 \geq 0$ nor $s_1 \geq 0$ binds, the Lagrange multipliers $\mu_0 = \mu_1 = 0$. From the first-order conditions for savings, $\lambda_0 = \lambda_1 = \lambda_2$. The Euler equations give the same consumption ratios as Model F. From the first-order condition for investment, $\lambda_0 = \lambda_2 f'(i)$, so $f'(i) = 1$ and $i = i^F$. The two models agree when both constraints are slack.

The constraint $s_1 \geq 0$ holds if and only if

$$s_1 = s_0 + y - c_1 \geq 0. \tag{1}$$

Substituting the unconstrained allocation and simplifying, this requires $W \geq \bar{W}$, where $\bar{W}$ is the (finite) threshold at which $s_1 = 0$ under the unconstrained allocation. The exact expression for $\bar{W}$ depends on parameters; its existence and finiteness follow from the fact that s_1 is linear in W with a positive slope.

Model C: Regime 1 (Gap-period constrained). Suppose $s_1 = 0$ binds ($\mu_1 > 0$) but $s_0 \geq 0$ remains slack ($\mu_0 = 0$). From $\mu_0 = 0$ and the first-order condition for savings:

$$\lambda_0 = \lambda_1 \quad \Longrightarrow \quad \frac{1}{c_0} = \frac{\beta}{c_1} \quad \Longrightarrow \quad c_1 = \beta c_0.$$

From $s_1 = 0$ and the date-2 budget: $c_2 = f(i) = Ai^\alpha$.

The date-1 budget with $s_1 = 0$ gives $c_1 = s_0 + y$, and the date-0 budget gives $s_0 = W - c_0 - i$, so $c_1 = W - c_0 - i + y$. Combining with $c_1 = \beta c_0$:

$$\beta c_0 = W - c_0 - i + y \quad \Longrightarrow \quad c_0 = \frac{W + y - i}{1 + \beta}.$$

The first-order condition for investment is $\lambda_0 = \lambda_2 f'(i)$, i.e.,

$$\frac{1}{c_0} = \frac{\beta^2}{c_2} \cdot f'(i) = \frac{\beta^2}{Ai^\alpha} \cdot \alpha A i^{\alpha-1} = \frac{\beta^2 \alpha}{i}.$$

Substituting $c_0 = (W + y - i)/(1 + \beta)$:

$$\frac{1 + \beta}{W + y - i} = \frac{\beta^2 \alpha}{i}.$$

Cross-multiplying:

$$i(1+\beta) = \beta^2\alpha(W+y-i) = \beta^2\alpha(W+y) - \beta^2\alpha\, i.$$

Collecting terms:

$$i(1+\beta+\beta^2\alpha) = \beta^2\alpha(W+y).$$

Therefore:

$$i^C = \frac{\beta^2\alpha}{1+\beta+\beta^2\alpha}\,(W+y).$$

This is a linear function of $W+y$. In contrast, $i^F = (\alpha A)^{1/(1-\alpha)}$ is independent of W and y.

Model C: Regime 2 (Doubly constrained). Both $s_0 = 0$ and $s_1 = 0$ bind. Then $c_0 = W - i$, $c_1 = y$, $c_2 = f(i) = Ai^\alpha$. The first-order condition for investment uses $\lambda_0 = \lambda_2 f'(i)$:

$$\frac{1}{W-i} = \frac{\beta^2\alpha}{i}.$$

Solving: $i = \beta^2\alpha\, W/(1+\beta^2\alpha)$. Investment is proportional to W but independent of y.

Summary. In all three regimes of Model C, the investment rule depends on the agent's wealth when the relevant constraint binds. In Model F, the investment rule is $i^F = (\alpha A)^{1/(1-\alpha)}$, which is independent of wealth. The qualitative divergence—from a technology condition to a wealth-dependent liquidity rule—is established. ■

A.2 Proof of Proposition 4.4 (KT Re-emergence)

Proof. We establish the existence of Kuhn-Tucker conditions at each commitment point and show that the multiplier is strictly positive on a set of positive measure.

Step 1: Structure of the commitment-point problem. Consider an agent at a commitment point ($s = \tau$) with state (a, k, z) where $a > \underline{a}$. By equation (44), the agent solves

$$\max_{i \geq 0}\; V\big(a - i,\, k + i_{\text{prev}},\, z;\, 0^+\big) \quad \text{subject to} \quad a - i \geq \underline{a}.$$

This is a finite-dimensional optimization problem over the single variable $i \in [0, a-\underline{a}]$. The objective function $i \mapsto V(a-i, k+i_{\text{prev}}, z; 0^+)$ is concave in i because V is concave in a (which follows from the concavity of u and the linearity of the budget constraint in a; see Achdou et al. (2022), Theorem 2). The constraint set $\{i : 0 \leq i \leq a - \underline{a}\}$ is convex and compact. Hence a unique maximizer i^* exists.

Step 2: Derivation of the KT conditions. The Lagrangian for the commitment-point problem is given by (47):

$$\mathcal{L}(i, \mu) = V\big(a - i,\, k + i_{\text{prev}},\, z;\, 0^+\big) + \mu\,(a - i - \underline{a}).$$

Since the objective is concave and continuously differentiable in i (by the smoothness of the flow-phase value function, which inherits regularity from the parabolic HJB equation (45)), and the constraint is linear, Slater's condition is satisfied whenever $a > \underline{a}$ (the point $i = 0$ is strictly feasible). Therefore the KKT conditions are necessary and sufficient for optimality:

$$-V_a\big(a - i^*,\, k + i_{\text{prev}},\, z;\, 0^+\big) + \mu = 0, \tag{48}$$

$$\mu \geq 0, \tag{49}$$

$$\mu\,(a - i^* - \underline{a}) = 0. \tag{50}$$

This establishes the first claim: the optimal investment at each commitment point satisfies complementary slackness conditions with a Kuhn-Tucker multiplier $\mu(a, k, z) \geq 0$.

Step 3: The buffer process during gestation. To show that $\mu > 0$ on a set of positive measure, we analyze the agent's liquidity position during the gestation interval. After committing investment i, the agent enters Phase 1 with liquid wealth $a - i$ and no investment expenditure. Define the buffer $b_t \equiv a_t - \underline{a}$, which measures the distance of liquid wealth from the borrowing limit. The buffer starts at $b_0 = a - i - \underline{a} \geq 0$ and evolves as

$$\mathrm{d}b_t = \big[r\,(b_t + \underline{a}) \cdot (k/k_t) + w\,z_t - c_t^*\big]\,\mathrm{d}t + \sigma_a\,\mathrm{d}W_t^a + \sigma_z(z_t)\,\mathrm{d}W_t^z,$$

where c_t^* is the optimal consumption during Phase 1 (characterized by the flow-phase FOC (43)). The borrowing constraint requires $b_t \geq 0$ for all $t \in [0, \tau]$.

To obtain a tractable bound, note that the buffer process is dominated by a drifted Brownian motion. Specifically, there exist constants $\mu_b \in \mathbb{R}$ and $\sigma_b > 0$ (depending on the state (a, k, z) and on the optimal consumption policy) such that

$$\Pr\Big(\min_{t\in[0,\tau]} b_t \leq 0 \;\Big|\; b_0\Big) \;\geq\; p(b_0, \tau),$$

where $p(b_0, \tau)$ is the first-passage probability for a Brownian motion with drift μ_b and volatility σ_b. By the reflection principle for Brownian motion with drift (see, e.g., Kushner and Dupuis (2001)), this probability admits the closed form

$$p(b_0, \tau) = \Phi\left(\frac{-b_0 + \mu_b \tau}{\sigma_b \sqrt{\tau}}\right) + \exp\left(\frac{-2\mu_b\, b_0}{\sigma_b^2}\right) \Phi\left(\frac{-b_0 - \mu_b \tau}{\sigma_b \sqrt{\tau}}\right), \tag{2}$$

where $\Phi(\cdot)$ is the standard normal CDF. The hitting probability (2) has the following properties:

(i) *Decreasing in* b_0: $\partial p / \partial b_0 < 0$, since a larger buffer reduces the probability of hitting the constraint.

(ii) *Increasing in* τ: $\partial p / \partial \tau > 0$, since a longer gestation period increases exposure.

(iii) *Vanishing as* $\tau \to 0$: for any fixed $b_0 > 0$, $p(b_0, \tau) \to 0$ as $\tau \to 0$, because $\Phi((-b_0 + \mu_b \tau)/(\sigma_b \sqrt{\tau})) \to \Phi(-\infty) = 0$ and the second term is bounded by $\Phi((-b_0 - \mu_b \tau)/(\sigma_b \sqrt{\tau})) \to 0$.

(iv) *Converging to one as* $b_0 \to 0^+$: for any fixed $\tau > 0$, $p(0^+, \tau) = 1$ when $\sigma_b > 0$, since a Brownian path starting at the boundary crosses it immediately.

Step 4: $\mu > 0$ on a set of positive measure. We now construct the set $\mathcal{S}$. Suppose $\tau > 0$ and $\sigma_a > 0$ (the case $\sigma_Z > 0$ is analogous, working through the induced volatility in prices). Consider any state (a, k, z) with $a > \underline{a}$. Define the unconstrained optimal investment $i^{\text{unc}}(a, k, z)$ as the solution to $V_a(a - i, k + i_{\text{prev}}, z; 0^+) = 0$ when it exists, or $i^{\text{unc}} = a - \underline{a}$ otherwise. The post-investment buffer at the unconstrained optimum is $b_0^{\text{unc}} = a - i^{\text{unc}} - \underline{a}$.

The commitment-point constraint (44) requires not only that $a - i \geq \underline{a}$ at the commitment point, but also that the agent can sustain $a_t \geq \underline{a}$ throughout $[0, \tau]$ under the

optimal Phase 1 consumption policy. When $\sigma_a > 0$ and $\tau > 0$, the hitting probability (2) implies that starting from any buffer $b_0 > 0$, there is a positive probability that the constraint binds during gestation. The value function V incorporates this risk: $V(a-i, k+i_{\text{prev}}, z; 0^+)$ already reflects the cost of the borrowing constraint potentially binding during the flow phase. In particular, $V_{aa} < 0$ strictly (the value function is strictly concave in a), and the marginal value of liquid wealth V_a is increasing in the commitment-period constraint tightness.

Define

$$\mathcal{S} = \big\{(a, k, z) : a > \underline{a},\ \ i^*(a, k, z) = a - \underline{a}\big\},$$

i.e., the set of states at which the borrowing constraint binds at the commitment point. For any $(a, k, z) \in \mathcal{S}$, the complementary slackness condition (50) gives $\mu = V_a(\underline{a}, k + i_{\text{prev}}, z; 0^+)$, and by (46) this is strictly positive.

It remains to show that $\mathcal{S}$ has positive measure under the stationary distribution. Property (iv) implies that for agents with a close to $\underline{a}$, the hitting probability is close to one, making the effective constraint very tight. Specifically, for any (k, z), define $\hat{a}(k, z) > \underline{a}$ as the asset level at which $i^{\text{unc}}(\hat{a}, k, z) = \hat{a} - \underline{a}$. By the strict concavity of V in a and the fact that $V_a(\underline{a}, k, z; 0^+) = +\infty$ while V_a is bounded at the unconstrained optimum, such $\hat{a}(k, z)$ exists. For all $a \in (\underline{a}, \hat{a}(k, z)]$, we have $(a, k, z) \in \mathcal{S}$. Since the stationary distribution of the Aiyagari-type model places positive mass on a neighborhood of $\underline{a}$ (the density of the stationary distribution is bounded below near the constraint; see Achdou et al. (2022)), the set $\mathcal{S}$ has positive measure under the ergodic distribution.

Step 5: The deterministic case. When $\sigma_a = \sigma_z = \sigma_Z = 0$, the buffer process b_t is deterministic. The borrowing constraint binds during gestation if and only if the deterministic trajectory $\{b_t\}_{t\in[0,\tau]}$ reaches zero. Define $t^* = \inf\{t \geq 0 : b_t = 0\}$. If $t^* \leq \tau$, the agent must reduce investment below the unconstrained optimum to ensure $b_t > 0$ throughout $[0, \tau]$, and the Kuhn-Tucker multiplier is strictly positive. The condition $t^* \leq \tau$ is determined entirely by b_0, the deterministic drift, and τ; no stochastic element is involved. This confirms that the commitment cost—and the resulting KT multiplier—arises from the temporal wedge $\tau > 0$, not from uncertainty. ■

A.3 Proof of Proposition 4.5 (Deterministic KT)

Proof. We prove the result constructively by exhibiting a deterministic economy in which $\mu > 0$ at the commitment point, and then establishing the general claim.

Step 1: Setup of the deterministic economy. Consider a deterministic economy ($\sigma_a = \sigma_z = \sigma_Z = 0$) with a single representative agent, CES production

$$F(K, L) = \left[\alpha\, K^{(\epsilon-1)/\epsilon} + (1-\alpha)\, L^{(\epsilon-1)/\epsilon}\right]^{\epsilon/(\epsilon-1)},$$

elasticity of substitution $\epsilon > 0$, depreciation rate $\delta > 0$, and time-to-build delay $\tau > 0$. The agent supplies one unit of labor inelastically ($L = 1$). Prices are given by marginal products: $r_t = F_K(K_t, 1) - \delta$ and $w_t = F_L(K_t, 1)$. Start the economy at $K_0 < K^*$, where K^* is the unique steady-state capital stock satisfying $F_K(K^*, 1) = \rho + \delta$.

On the transition path to K^*, capital is below steady state, so the interest rate $r_t = F_K(K_t, 1) - \delta$ is above its steady-state level and is *declining* as $K_t \uparrow K^*$ (since $F_{KK} < 0$). Meanwhile, the wage $w_t = F_L(K_t, 1)$ is below its steady-state level and is *rising* (since $F_{KL} > 0$ with CES and $\epsilon > 0$). Hence $\dot{r}_t < 0$ and $\dot{w}_t > 0$ along the transition.

Step 2: The agent's problem during gestation. Suppose the agent arrives at a commitment point at date t_0 with liquid wealth a_0 and commits investment $i \geq 0$. The investment expenditure i is deducted immediately; the capital augmentation occurs at date $t_0 + \tau$. During the gestation interval $[t_0, t_0 + \tau]$, the agent's liquid wealth evolves deterministically under optimal consumption:

$$\dot{a}_t = r_t\, a_t + w_t - c_t^*, \qquad a_{t_0} = a_0 - i, \qquad a_t \geq \underline{a} \;\; \forall\, t \in [t_0, t_0 + \tau], \tag{3}$$

where c_t^* is the optimal consumption path during Phase 1, given by the deterministic Euler equation

$$\dot{c}_t / c_t = (r_t - \rho)/\gamma,$$

with $\gamma > 0$ being the coefficient of relative risk aversion. On the transition path with $r_t > \rho$ (when K_0 is sufficiently below K^*), consumption is rising: $\dot{c}_t > 0$.

Step 3: Constructing a state with $\mu > 0$. Fix any (k, z) and consider the unconstrained optimum i^{unc}, defined as the investment level that maximizes the continuation value absent the gestation-period constraint. The post-investment liquid wealth is $a_0 - i^{\text{unc}}$.

We claim that for a_0 sufficiently close to $\underline{a}+i^{\text{unc}}$ from above, the deterministic path $\{a_t\}_{t\in[t_0,t_0+\tau]}$ starting from $a_{t_0} = a_0 - i^{\text{unc}}$ violates the borrowing constraint $a_t \geq \underline{a}$ at some interior date $t \in (t_0, t_0 + \tau)$, even though $a_{t_0} = a_0 - i^{\text{unc}} > \underline{a}$ at the commitment point itself.

To see this, note that the deterministic evolution (3) can generate a *U-shaped* path of liquid wealth during gestation. This occurs because: (i) the agent's income flow $r_t a_t + w_t$ may be low relative to the rising consumption path early in the gestation interval (consumption rises since $r_t > \rho$), and (ii) the declining interest rate reduces the return on savings over the interval. Specifically, when r_t is declining, the agent's liquid assets earn a diminishing return, while the optimal consumption plan (based on the Euler equation) prescribes increasing consumption. For a small initial buffer $b_0 = a_{t_0} - \underline{a} > 0$, there exists $t^* \in (t_0, t_0 + \tau)$ at which $a_{t^*} < \underline{a}$.

Formally, define the minimum liquid wealth along the unconstrained path:

$$a^{\min}(a_0, i) \equiv \min_{t\in[t_0,t_0+\tau]} a_t(a_0 - i),$$

where $a_t(a_0 - i)$ solves (3) with initial condition $a_{t_0} = a_0 - i$. The function $a^{\min}$ is continuous in (a_0, i) and strictly increasing in $a_0 - i$ (since a higher initial wealth uniformly raises the path). At the unconstrained optimum:

- If $a^{\min}(a_0, i^{\text{unc}}) \geq \underline{a}$, the gestation-period constraint does not bind and $\mu = 0$.
- If $a^{\min}(a_0, i^{\text{unc}}) < \underline{a}$, the agent must reduce investment below i^{unc} to satisfy the constraint, and $\mu > 0$.

The second case obtains for initial conditions with a_0 close to $\underline{a} + i^{\text{unc}}$. To verify existence, consider the limiting case $a_0 = \underline{a} + i^{\text{unc}}$, so that $a_{t_0} = \underline{a}$ and $b_0 = 0$. Then $a^{\min} = \underline{a}$ only if the path never falls below $\underline{a}$, which requires

$$r_t\,\underline{a} + w_t \geq c_t^* \qquad \forall\, t \in [t_0, t_0 + \tau].$$

Since $\dot{r}_t < 0$ and $\dot{c}_t > 0$ along the transition, there generically exists $t' \in (t_0, t_0 + \tau)$

at which $r_{t'}\,\underline{a} + w_{t'} < c^*_{t'}$, causing $\dot{a}_{t'} < 0$. Starting from $a_{t_0} = \underline{a}$, this yields $a_{t'} < \underline{a}$. By continuity, the same violation occurs for a_0 in a neighborhood of $\underline{a} + i^{\text{unc}}$.

Step 4: Positive measure. The constrained optimum $i^*(a_0)$ is determined by the binding path constraint:

$$i^*(a_0) = \max\big\{i \geq 0 : a^{\min}(a_0, i) \geq \underline{a}\big\}.$$

At this constrained optimum, $a^{\min}(a_0, i^*(a_0)) = \underline{a}$, and the Kuhn-Tucker multiplier on the path constraint is

$$\mu(a_0) = V_a(a_0 - i^*, k + i_{\text{prev}}, z; 0^+) > 0,$$

where the strict inequality follows from the boundary condition (46) and the concavity of V.

The set of initial conditions for which the path constraint binds is

$$\mathcal{S}_{\text{det}} = \big\{(a, k, z) : a > \underline{a},\ \ a^{\min}(a, i^{\text{unc}}) < \underline{a}\big\}.$$

By the construction above, $\mathcal{S}_{\text{det}}$ contains a neighborhood of $\{a : a \in (\underline{a}, \underline{a} + i^{\text{unc}} + \varepsilon)\}$ for some $\varepsilon > 0$, and therefore has positive Lebesgue measure in the state space.

Step 5: Generality. The construction above uses the CES specification for concreteness, but the argument requires only three ingredients: (i) $\tau > 0$ so that the gestation interval is non-degenerate; (ii) the economy is on a transition path so that prices change during $[t_0, t_0 + \tau]$ ($\dot{r}_t \neq 0$ or $\dot{w}_t \neq 0$); and (iii) the borrowing limit $\underline{a}$ is finite. Under these conditions, the temporal mismatch between the agent's income profile and consumption profile during gestation generates states at which the deterministic path of liquid wealth violates the borrowing constraint, making $\mu > 0$. No uncertainty is required; the Kuhn-Tucker multiplier arises purely from the commitment cost. ■

A.4 Proof of Proposition 4.6 (Convergence and Interpolation)

Proof. The time-to-build parameter $\tau \geq 0$ indexes a family of continuous-time models $\{\mathcal{M}(\tau)\}_{\tau \geq 0}$. We establish each part of the proposition in turn.

Part (i): Convergence as $\tau \to 0$ to the standard CT model. We use the stability theory of viscosity solutions following the framework of Barles and Souganidis (1991).

Step 1: The family of HJB systems. For each $\tau > 0$, the value function $V^\tau(a, k, z; s)$ satisfies the flow-phase HJB equation (45) on the domain $(a, k, z) \in (\underline{a}, \infty) \times \mathbb{R}_+ \times \mathcal{Z}$ for $s \in (0, \tau)$, with the commitment-point boundary condition

$$V^\tau(a, k, z; \tau) = \max_{i \in [0, a - \underline{a}]} V^\tau\big(a - i, k + i_{\text{prev}}, z; 0^+\big)$$

at $s = \tau$, and the state-constraint boundary condition $V_a^\tau \to +\infty$ as $a \to \underline{a}^+$. The standard CT model corresponds to $\tau = 0$, with value function $V^0(a, k, z)$ satisfying the HJB equation the standard CT HJB equation with continuous investment control.

Step 2: Uniform bounds. For each $\tau > 0$, the value function V^τ is bounded above by the value function of an agent who faces no borrowing constraint (which is finite under standard growth conditions on u), and bounded below by the value function of an agent who never invests ($i \equiv 0$). These bounds are independent of τ, so the family $\{V^\tau\}_{\tau > 0}$ is uniformly bounded on compact subsets of the state space.

Moreover, V^τ is Lipschitz continuous in a on $[\underline{a} + \varepsilon, \bar{a}]$ for any $\varepsilon > 0$ and $\bar{a} < \infty$, with a Lipschitz constant that depends on ε and the model parameters but not on τ (this follows from the uniform concavity of u and the linear budget constraint). The family $\{V^\tau\}$ is therefore equicontinuous on compact subsets of $\{a > \underline{a}\}$.

Step 3: Convergence of the commitment-point condition. As $\tau \to 0$, the timer domain $(0, \tau)$ shrinks to a point. Over the flow phase of length τ, the agent optimizes consumption but not investment. The value of the flow phase, relative to the value at the commitment point, satisfies

$$V^\tau(a, k, z; 0^+) = V^\tau(a, k, z; \tau) + O(\tau),$$

where the $O(\tau)$ term captures the flow utility and the evolution of states over the

interval of length τ. The commitment-point condition becomes

$$V^\tau(a,k,z;\tau) = \max_{i\in[0,a-\underline{a}]} \big[V^\tau(a-i,k+i_{\text{prev}},z;\tau) + O(\tau)\big].$$

As $\tau \to 0$, the commitment points $\{0,\tau,2\tau,\ldots\}$ become dense in $\mathbb{R}_+$, and the agent effectively chooses investment at every instant. The constraint $a-i \geq \underline{a}$ becomes the standard state-constraint boundary condition of the CT model.

Step 4: Applying the Barles-Souganidis framework. We verify the three conditions of the Barles-Souganidis convergence theorem:

(a) *Monotonicity*: The flow-phase HJB operator is a second-order uniformly elliptic operator (when $\sigma_a > 0$). The comparison principle for the HJB equation (45) holds: if $V_1 \leq V_2$ on the boundary of the domain, then $V_1 \leq V_2$ on the interior. The commitment-point operator preserves the ordering because V^τ is increasing in the continuation value. Hence the combined scheme (flow HJB + commitment-point boundary condition) is monotone.

(b) *Stability*: The uniform bounds from Step 2 provide the required stability: the family $\{V^\tau\}_{\tau>0}$ is uniformly bounded on compact sets.

(c) *Consistency*: The flow-phase HJB equation (45) is the same for all τ; only the domain $s \in (0,\tau)$ changes. As $\tau \to 0$, the flow phase vanishes and the operator reduces to the standard CT HJB equation the standard CT HJB equation with continuous investment control. The consistency of this reduction follows from the $O(\tau)$ expansion in Step 3.

By the Barles-Souganidis theorem, any subsequential limit of V^τ as $\tau \to 0$ is a viscosity solution of the standard CT HJB equation. By the uniqueness of viscosity solutions for the CT model (which holds under the standard assumptions; see Achdou et al. (2022)), the limit is unique: $V^\tau \to V^0$ uniformly on compact subsets of $\{a > \underline{a}\}$.

Step 5: Convergence of policies. The optimal consumption policy c^τ is determined by $u'(c^\tau) = V_a^\tau$. Since $V^\tau \to V^0$ uniformly on compact sets and V^τ is strictly concave in a, the derivatives V_a^τ converge to V_a^0 on the interior, and hence $c^\tau \to c^0$. The optimal investment policy i^τ at commitment points converges to the continuous investment policy i^0 of the CT model because the commitment-point condition (44) reduces to the continuous FOC the continuous-time FOC in the limit.

The Kuhn-Tucker multiplier $\mu^\tau \to 0$ uniformly on compact subsets of $\{a > \underline{a}\}$: by property (iii) of the hitting probability (2), for any $a > \underline{a}$, the probability of the borrowing constraint binding during gestation vanishes as $\tau \to 0$, so the effective constraint on investment becomes slack and $\mu^\tau \to 0$.

Part (ii): Convergence as $\tau \to \Delta t$ to the DT model. *Step 1: Identification with the DT Bellman equation.* When $\tau = \Delta t$, the commitment period equals the discrete-time period length. The agent commits investment at dates $0, \Delta t, 2\Delta t, \ldots$ and chooses consumption continuously during each interval. The Phase 1 value function over $[0, \Delta t]$ aggregates the flow utility and the evolution of states over one full period.

Define the *period value function*

$$W(a, k, z) \equiv V^{\Delta t}(a, k, z; 0^+),$$

which gives the value at the start of a flow phase (immediately after a commitment point). The commitment-point condition is

$$V^{\Delta t}(a, k, z; \Delta t) = \max_{i \in [0, a-\underline{a}]} W(a - i, k + i_{\text{prev}}, z).$$

The function $W(a, k, z)$ itself is determined by solving the flow-phase HJB (45) over $s \in (0, \Delta t)$ with terminal condition $V^{\Delta t}(a, k, z; \Delta t)$ at $s = \Delta t$. This yields a mapping from the terminal value $V^{\Delta t}(\cdot; \Delta t)$ to the initial value $W(\cdot) = V^{\Delta t}(\cdot; 0^+)$, which is precisely the one-period Bellman operator of the discrete-time model.

Step 2: Recovery of the DT Euler equation. The flow-phase HJB over $[0, \Delta t]$ integrates the Euler equation $u'(c_t) = V_a^{\Delta t}(a_t, k_t, z_t; t)$ along the optimal path. At the commitment point $s = \Delta t$, the optimality condition yields

$$u'(c_0) = \mathbb{E}_0\big[e^{-\rho\,\Delta t}\, V_a^{\Delta t}(a_{\Delta t}, k_{\Delta t}, z_{\Delta t}; \Delta t)\big],$$

where the expectation is over the evolution of (a, k, z) during $[0, \Delta t]$. Writing $\beta \equiv e^{-\rho\,\Delta t}$ and noting that $V_a^{\Delta t}(a, k, z; \Delta t) = u'(c_{\Delta t}) + \mu_{\Delta t}$ at the next commitment point (by the envelope theorem applied to the commitment-point problem), we obtain

$$u'(c_0) = \beta\,\mathbb{E}_0\big[u'(c_{\Delta t})\big] + \beta\,\mathbb{E}_0[\mu_{\Delta t}],$$

which is the standard DT Euler equation with Kuhn-Tucker multiplier.

Step 3: KT structure. The commitment-point condition at $s = \Delta t$ is

$$\max_{i \in [0, a-\underline{a}]} W(a - i, k + i_{\text{prev}}, z), \qquad \mu_{\Delta t} \geq 0, \qquad \mu_{\Delta t}(a - i^* - \underline{a}) = 0.$$

This is exactly the DT investment optimality condition with the Kuhn-Tucker multiplier appearing at each period boundary, as in the standard DT Bellman equation the standard DT Bellman equation. Hence $\mathcal{M}(\Delta t)$ reproduces the DT model at period length Δt.

Part (iii): Continuous interpolation for $\tau \in (0, \Delta t)$. *Step 1: Continuity of the value function in τ.* For each fixed (a, k, z) with $a > \underline{a}$, the map $\tau \mapsto V^\tau(a, k, z)$ is continuous. This follows from the stability of viscosity solutions under perturbation of the domain: as τ varies continuously, the flow-phase domain $(0, \tau)$ and the commitment-point boundary condition change continuously. The standard stability result for viscosity solutions of parabolic PDEs (see Crandall et al. (1992), Theorem 8.3) implies that V^τ depends continuously on τ in the locally uniform topology.

Step 2: Continuity of policies and distributions. The optimal consumption policy $c^\tau = (u')^{-1}(V_a^\tau)$ inherits continuity in τ from the continuity of V^τ. The optimal investment policy i^τ at commitment points is determined by the commitment-point FOC, which is continuous in τ via the maximum theorem (applied to the commitment-point problem, which has a continuous objective and a continuous constraint set). The stationary distribution μ^τ satisfies the Kolmogorov forward equation, whose drift and diffusion coefficients depend continuously on the policy functions (c^τ, i^τ). By the continuous dependence of stationary distributions on drift and diffusion (see, e.g., Bogachev et al. (2015), Chapter 6), the map $\tau \mapsto \mu^\tau$ is continuous in the weak topology.

Step 3: The interpolation property. Combining Parts (i)–(iii): the family $\{V^\tau, c^\tau, i^\tau, \mu^\tau\}_{\tau \in [0, \Delta t]}$ varies continuously in τ, with the endpoints $\tau = 0$ and $\tau = \Delta t$ recovering the standard CT and DT models, respectively. For intermediate values $\tau \in (0, \Delta t)$, the model $\mathcal{M}(\tau)$ provides a continuous interpolation between the two formulations. The Kuhn-Tucker multiplier μ^τ increases continuously from 0 (at $\tau = 0$, the CT limit) to the DT multiplier (at $\tau = \Delta t$), parameterized by the structural commitment lag τ. ■

A.5 Proof of Proposition 4.7 (Precautionary Under-Investment)

Proof. We establish each claim using monotone comparative statics arguments based on the lattice-theoretic framework of Topkis (1998) and Milgrom and Shannon (1994).

Step 1: The commitment-point objective and its cross-partial. At a commitment point, the agent with state (a, k, z) solves

$$\max_{i \in [0, a-\underline{a}]} \Pi(i, \tau) \equiv V^{\tau}\big(a - i,\ k + i_{\text{prev}},\ z;\ 0^{+}\big),$$

where V^{τ} denotes the Phase 1 value function under commitment delay τ. The key structural property is the *submodularity* of Π in (i, τ): a higher τ reduces the marginal return to investment i.

We establish $\Pi_{i\tau} \leq 0$ (the cross-partial condition for submodularity) through two channels.

Channel A: The marginal value of liquid wealth rises with τ. The derivative of Π with respect to i is $\Pi_i = -V_a^{\tau}(a - i, k + i_{\text{prev}}, z; 0^{+})$, where V_a^{τ} is the marginal value of post-commitment liquid wealth. We claim that V_a^{τ} is increasing in τ for each fixed (a, k, z) with a near $\underline{a}$.

To see this, consider two gestation intervals $\tau_2 > \tau_1 > 0$. The agent under τ_2 must sustain consumption from liquid wealth over a longer interval $[0, \tau_2]$ without receiving any investment return. During the additional interval $[\tau_1, \tau_2]$, the agent's liquid wealth continues to be drawn down by consumption, and the borrowing constraint $a_t \geq \underline{a}$ must be maintained over a larger set. The value of an additional unit of liquid wealth at the start of the flow phase is therefore (weakly) higher under τ_2 than under τ_1:

$$V_a^{\tau_2}(a, k, z; 0^{+}) \geq V_a^{\tau_1}(a, k, z; 0^{+}) \qquad \text{for all } (a, k, z).$$

This inequality can be established formally by a comparison argument on the flow-phase HJB equation (45). Consider the flow-phase PDE as a terminal-value problem on $s \in (0, \tau)$. The terminal condition is the commitment-point value. The solution operator of the parabolic PDE is a contraction in the sup-norm and preserves the ordering of gradients in a: a longer integration interval (larger τ) produces a value function with weakly steeper marginal value in a, because the agent faces a more constrained problem (longer period without investment income). This is a standard

result in the theory of parabolic PDEs with state constraints (see Lions (1983), Chapter IV).

Consequently, $\Pi_{i\tau} = -\partial V_a^\tau / \partial \tau \leq 0$, confirming submodularity.

Channel B: The effective constraint tightens with τ. When the borrowing constraint $a_t \geq \underline{a}$ is at risk of binding during the gestation interval, the agent faces an effective upper bound on investment: $i \leq \bar{\imath}(a, k, z; \tau)$, where $\bar{\imath}$ is the maximum investment consistent with maintaining $a_t \geq \underline{a}$ throughout $[0, \tau]$. By property (ii) of the hitting probability (2), a longer gestation interval increases the probability of hitting the constraint, which (weakly) reduces $\bar{\imath}$:

$$\bar{\imath}(a, k, z; \tau_2) \leq \bar{\imath}(a, k, z; \tau_1) \qquad \text{for } \tau_2 > \tau_1.$$

The effective constraint set $[0, \bar{\imath}(a, k, z; \tau)]$ is therefore shrinking in τ, which further reduces the optimal investment through the tightening of the Kuhn-Tucker multiplier.

Step 2: Monotone comparative statics for $i^*(\tau)$. By Topkis's monotonicity theorem (Topkis (1998), Theorem 2.8.1), if $\Pi(i, \tau)$ is submodular in (i, τ) and the constraint set $[0, \bar{\imath}(a, k, z; \tau)]$ is decreasing in τ (in the strong set order), then the optimal investment $i^*(a, k, z; \tau) = \arg\max_{i \in [0, \bar{\imath}]} \Pi(i, \tau)$ is decreasing in τ.

We have established both conditions: $\Pi_{i\tau} \leq 0$ (Channel A) and $\bar{\imath}$ decreasing in τ (Channel B). Therefore $i^*(a, k, z; \tau)$ is weakly decreasing in τ for each (a, k, z).

The inequality is strict for agents in the set $\mathcal{S}$ identified in the proof of Proposition 4.4 (Appendix A.2): for these agents, either the submodularity is strict (V_a^τ is strictly increasing in τ because the constraint is binding with positive probability) or the effective constraint is strictly tighter ($\bar{\imath}$ is strictly decreasing in τ). This yields

$$i^*(a, k, z; \tau_2) < i^*(a, k, z; \tau_1) \qquad \text{for all } (a, k, z) \in \mathcal{S} \text{ and } \tau_2 > \tau_1.$$

Step 3: Part (i) — Lower average investment. Since $i^*(a, k, z; \tau)$ is weakly decreasing in τ for all (a, k, z) and strictly decreasing on $\mathcal{S}$ (which has positive measure under the stationary distribution, as shown in Proposition 4.4), the average investment under the stationary distribution is strictly lower:

$$\mathbb{E}_{\mu^\tau}[i^*(\tau)] \leq \mathbb{E}_{\mu^0}[i^*(0)].$$

To obtain the strict inequality $\mathbb{E}[i^*(\tau)] < \mathbb{E}[i^*(0)]$, note that the stationary distribution μ^τ places positive mass on the set $\mathcal{S}$ (by the argument in Proposition 4.4, Step 4). On $\mathcal{S}$, $i^*(\tau) < i^*(0)$ strictly. Hence the integral is strictly smaller.

Step 4: Part (ii) — Higher average liquid asset holdings. The reduction in investment redirects resources toward liquid asset accumulation. Formally, in the stationary distribution the budget constraint implies

$$\mathbb{E}_{\mu^\tau}[c^\tau + i^\tau] = \mathbb{E}_{\mu^\tau}[r\,k + w\,z].$$

Since i^τ is lower for agents in $\mathcal{S}$ and the right-hand side (aggregate income) adjusts through general equilibrium, the reduced investment leads to higher liquid asset holdings in steady state. More precisely, consider partial equilibrium (fixed prices). Lower investment at each commitment point implies higher post-commitment liquid wealth $a - i$. The flow-phase dynamics (40) then imply higher liquid wealth throughout the flow phase. In the stationary distribution, this shifts mass rightward in a:

$$\mathbb{E}_{\mu^\tau}[a] > \mathbb{E}_{\mu^0}[a] \qquad \text{for } \tau > 0.$$

The argument extends to general equilibrium: lower aggregate investment reduces the steady-state capital stock, raising r and lowering w, which amplifies the incentive to hold liquid assets (a standard general-equilibrium feedback in Aiyagari-type models).

Step 5: Part (iii) — Steeper investment policy near $\underline{a}$. We show that the investment policy $i^*(a, k, z; \tau)$ has a steeper gradient in a near $\underline{a}$ when $\tau > 0$. The intuition is that the commitment cost is most severe for asset-poor agents, creating a sharp drop-off in investment as $a \to \underline{a}^+$.

Formally, consider the first-order condition at the commitment point when the constraint binds: $V_a^\tau(a - i^*, k + i_{\text{prev}}, z; 0^+) = \mu^\tau$. Differentiating the optimality condition $V_a^\tau(a - i^*, k + i_{\text{prev}}, z; 0^+) = \mu^\tau(a, k, z; \tau)$ with respect to a by the implicit function theorem:

$$V_{aa}^\tau\left(1 - \frac{\partial i^*}{\partial a}\right) = \frac{\partial \mu^\tau}{\partial a} \qquad \Longrightarrow \qquad \frac{\partial i^*}{\partial a} = 1 - \frac{\partial \mu^\tau / \partial a}{V_{aa}^\tau}.$$

Since $V_{aa}^\tau < 0$ (strict concavity) and $\partial \mu^\tau / \partial a < 0$ (the multiplier decreases as the

agent becomes wealthier), the ratio $(\partial\mu^\tau/\partial a)/V^\tau_{aa} > 0$, so $\partial i^*/\partial a < 1$. Near $a = \underline{a}$, the constraint is tightly binding (μ^τ is large and changes rapidly with a), while $|V^\tau_{aa}|$ is large (reflecting the high curvature of the value function near the constraint). The gradient $\partial i^*/\partial a$ is therefore large near $\underline{a}$: a small increase in wealth translates almost one-for-one into higher investment, as the relaxation of the binding constraint allows most of the additional wealth to be invested.

When $\tau > 0$, this gradient is strictly larger than under $\tau = 0$ because: (i) the commitment cost introduces an additional source of constraint tightness (the gestation-period constraint, not just the point constraint), and (ii) the sensitivity of μ^τ to a is higher near $\underline{a}$ when $\tau > 0$ (since the hitting probability (2) is most sensitive to the initial buffer $b_0 = a - i - \underline{a}$ when b_0 is small). Hence $|\partial i^*/\partial a|$ near $a = \underline{a}$ is increasing in τ.

Step 6: Monotonicity in τ. All three effects are monotonically increasing in τ by the monotone comparative statics argument of Step 2. As τ increases: (a) the submodularity of Π in (i, τ) implies that $i^*(\tau)$ is decreasing in τ, so the investment reduction grows; (b) the redirected resources increase liquid asset holdings further; and (c) the steepening of the investment policy near $\underline{a}$ intensifies because the hitting probability (2) is increasing in τ, raising the sensitivity of the Kuhn-Tucker multiplier to the agent's wealth position. The aggregate quantities inherit this monotonicity: $I(\tau) = \int i^*(a, k, z; \tau)\, \mathrm{d}\mu^\tau(a, k, z)$ is decreasing in τ and $\mathbb{E}_{\mu^\tau}[a]$ is increasing in τ. ■

B Computational Details

This appendix describes the numerical methods used to solve the commitment model of Section 5. The algorithm combines finite difference methods for the HJB equation (Phase 1), constrained optimization at commitment points (Phase 2), a Kolmogorov forward equation for the stationary distribution, and an outer loop for general equilibrium.

B.1 Grid Specification

The individual state is (a, k, z, s): liquid wealth a, installed capital k, idiosyncratic productivity z, and timer s (time remaining until the current investment project is

completed).

Liquid wealth a. We discretize $a \in [\underline{a}, \bar{a}]$ on a non-uniform grid with $N_a = 200$ points. The grid is concentrated near the borrowing limit $\underline{a} = 0$, where the value function has the greatest curvature. Specifically, we use a power-spaced grid:

$$a_j = \underline{a} + (\bar{a} - \underline{a}) \left(\frac{j}{N_a - 1} \right)^{\xi}, \qquad j = 0, 1, \ldots, N_a - 1,$$

with $\xi = 3$, so that approximately half the grid points lie in the bottom quartile of the wealth range. The upper bound $\bar{a}$ is set large enough that the value function is approximately linear at the boundary (verified ex post).

Installed capital k. We discretize $k \in [0, \bar{k}]$ on a uniform grid with $N_k = 50$ points. The upper bound $\bar{k}$ is chosen to exceed the steady-state capital stock by a wide margin.

Productivity z. The Ornstein-Uhlenbeck process for $\log z$ is discretized using the method of Achdou and Capuzzo-Dolcetta (2010). We use $N_z = 7$ grid points spanning ± 3 standard deviations of the stationary distribution.

Timer s. For each value of $\tau > 0$, the timer variable $s \in [0, \tau]$ is discretized on a uniform grid with N_s points, where $N_s = 4\tau/\Delta s$ and $\Delta s = 0.25$ (one quarter). For $\tau = 1$, $N_s = 4$; for $\tau = 4$, $N_s = 16$. The timer counts down from τ (project initiation) to 0 (project completion).

B.2 Phase 1: HJB Equation

For $s > 0$, the agent is in the gestation phase and chooses only consumption. The HJB equation is

$$\rho V(a, k, z, s) = \max_{c \geq 0} \left\{ u(c) + V_a(r\,a + wz - c) + V_k(-\delta k) + \mathcal{L}_z V + V_s \cdot (-1) \right\}, \quad (4)$$

where V_a, V_k, and V_s denote partial derivatives, $\mathcal{L}_z$ is the infinitesimal generator of the z process, and the $V_s \cdot (-1)$ term reflects the deterministic countdown of the timer ($\dot{s} = -1$).

The first-order condition for consumption gives $c^* = (u')^{-1}(V_a)$. With CRRA utility $u(c) = c^{1-\gamma}/(1-\gamma)$, this is $c^* = V_a^{-1/\gamma}$.

Finite difference scheme. We use an upwind finite difference scheme. The drift of a is $\mu_a = r\,a + wz - c$. Define forward and backward differences:

$$(V_a)_j^+ = \frac{V_{j+1} - V_j}{\Delta a_j^+}, \qquad (V_a)_j^- = \frac{V_j - V_{j-1}}{\Delta a_j^-},$$

where $\Delta a_j^+ = a_{j+1} - a_j$ and $\Delta a_j^- = a_j - a_{j-1}$. The upwind scheme selects the forward difference when the drift is positive and the backward difference when the drift is negative:

$$V_a \approx (V_a)_j^+ \cdot \mathbf{1}\{\mu_a > 0\} + (V_a)_j^- \cdot \mathbf{1}\{\mu_a < 0\}.$$

This choice ensures monotonicity of the scheme, which is required for convergence to the viscosity solution (Barles and Souganidis, 1991).

Boundary condition at $a = \underline{a}$. At the borrowing constraint, the state-constraint boundary condition requires non-negative drift: $r\,\underline{a} + wz - c \geq 0$. This pins down the consumption policy at the boundary: $c(\underline{a}, k, z, s) = r\,\underline{a} + wz$.

Backward stepping in s. The timer dimension is solved by backward stepping from $s = 0$ (the terminal condition, where the Phase 2 value is known) to $s = \tau$ (project initiation). At each step, the HJB equation (4) is solved for the consumption policy and value function on the (a, k, z) grid, using the value at the next timer step as the continuation value for the V_s term:

$$V_s(a, k, z, s_n) \approx \frac{V(a, k, z, s_{n-1}) - V(a, k, z, s_n)}{\Delta s}.$$

This is an implicit scheme in the s dimension, which is unconditionally stable.

B.3 Phase 2: Commitment Optimization

At $s = 0$, the current investment project is completed: capital k is delivered and becomes productive. The agent then chooses a new investment level $i \geq 0$, which sets

the timer to $s = \tau$ and reduces liquid wealth by i. The Phase 2 problem is

$$V(a, k, z, 0) = \max_{i \geq 0} \tilde{V}(a - i, k + i, z, \tau), \tag{5}$$

subject to the constraint that $a - i \geq \underline{a} + \bar{b}(z, \tau)$, where $\bar{b}(z, \tau)$ is the minimum buffer required to sustain the borrowing constraint through the gestation interval. In practice, $\bar{b}$ is computed from the Phase 1 solution: it is the smallest initial buffer such that the optimal consumption path during Phase 1 does not drive liquid wealth below $\underline{a}$.

The optimization (5) is solved at each grid point (a, k, z) by a one-dimensional search over $i \in [0, a - \underline{a} - \bar{b}(z, \tau)]$. The objective is evaluated by interpolating the Phase 1 value function $\tilde{V}$ at the post-commitment state $(a - i, k + i, z, \tau)$. We use golden section search combined with grid search to handle potential non-concavities.

The Kuhn-Tucker multiplier is computed as the shadow value of relaxing the upper bound on i:

$$\mu(a, k, z) = \max \left\{ -\tilde{V}_a(a - i^*, k + i^*, z, \tau) + \tilde{V}_k(a - i^*, k + i^*, z, \tau),\ 0 \right\},$$

where i^* is the constrained optimum. When $\mu > 0$, the agent is constrained: she would invest more if the commitment-period borrowing constraint were relaxed.

B.4 Kolmogorov Forward Equation

The stationary distribution $\mu(a, k, z, s)$ satisfies the Kolmogorov forward (KF) equation, which is the adjoint of the infinitesimal generator associated with the individual dynamics. In Phase 1 ($s > 0$), the KF equation is

$$0 = -\frac{\partial}{\partial a}[\mu_a(a, k, z, s) \cdot g] - \frac{\partial}{\partial k}[(-\delta k) \cdot g] + \mathcal{L}_z^* g + \frac{\partial g}{\partial s}, \tag{6}$$

where $g = g(a, k, z, s)$ is the density, $\mu_a = r\,a + wz - c^*(a, k, z, s)$ is the drift of liquid wealth under the optimal policy, and $\mathcal{L}_z^*$ is the adjoint of the generator of the z process.

The timer dimension is solved by forward stepping from $s = \tau$ (project initiation) to $s = 0$ (project completion). At $s = 0$, the mass of agents completing projects is redistributed across the (a, k, z) grid according to the Phase 2 investment policy: an

agent at $(a, k, z, 0)$ who invests $i^*(a, k, z)$ transitions to $(a - i^*, k + i^*, z, \tau)$.

The discretized KF equation uses the adjoint of the finite difference operator from the HJB equation, which ensures that the scheme is mass-conserving. The stationary distribution is computed by iterating the KF equation to convergence, starting from a uniform initial distribution, until $\|g^{n+1} - g^n\|_1 < 10^{-10}$.

B.5 General Equilibrium

The outer loop solves for the equilibrium interest rate r such that the capital market clears:

$$K(r) \equiv \int k \, \mathrm{d}\mu(a, k, z, s; r) = K^{\text{demand}}(r), \tag{7}$$

where $K^{\text{demand}}(r)$ is the capital demand from the firm's first-order condition, $r = \alpha A K^{\alpha-1} L^{1-\alpha} - \delta$. Given r, the wage is $w = (1 - \alpha) A K^{\alpha} L^{-\alpha}$.

The algorithm is:

1. Guess an interest rate $r^{(0)}$.
2. Given $r^{(n)}$, compute the wage $w^{(n)}$ from the firm's first-order condition.
3. Solve the Phase 1 HJB equation (backward in s, iterating to convergence in the (a, k, z) dimensions).
4. Solve the Phase 2 commitment optimization at $s = 0$.
5. Compute the stationary distribution from the KF equation.
6. Compute the aggregate capital supply $K^{(n)} = \int k \, \mathrm{d}\mu^{(n)}$.
7. Check the market clearing condition. If $|K^{(n)} - K^{\text{demand}}(r^{(n)})| < \varepsilon$, stop. Otherwise, update $r^{(n+1)}$ using bisection and return to step 2.

The tolerance is $\varepsilon = 10^{-6}$. The bisection interval for r is initialized using the analytical bounds $r \in (-\delta, \rho)$, which follow from the requirement that the steady-state capital stock is positive and finite.

B.6 Convergence Criteria and Verification

HJB convergence. The Phase 1 HJB equation is solved by implicit iteration. At each timer step s_n, the discretized HJB equation is a system of nonlinear equations in $V(a_j, k_l, z_m, s_n)$, which is solved by Howard's policy iteration: given a policy c, the value function is updated by solving the linear system implied by the HJB equation; given the updated value function, the policy is updated via the first-order condition. The inner loop converges when $\|V^{(p+1)} - V^{(p)}\|_\infty < 10^{-8}$.

KF convergence. The stationary distribution is computed by forward iteration of the KF equation. Convergence is declared when $\|g^{(n+1)} - g^{(n)}\|_1 < 10^{-10}$. We verify that the distribution integrates to one and that the implied aggregate statistics (mean wealth, fraction constrained) are stable across iterations.

Grid sensitivity. We verify that the results are robust to grid refinement by solving the model on coarser ($N_a = 100$) and finer ($N_a = 400$) grids. The equilibrium interest rate and aggregate capital change by less than 0.1% across these grid sizes, confirming that the baseline grid provides adequate resolution.

Discrete-time benchmark. The standard DT model is solved using value function iteration on the same (a, z) grid (without the k and s dimensions). The Euler equation with Kuhn-Tucker conditions is enforced using the endogenous grid method. The DT model serves as an independent check: the CT commitment model with $\tau = \Delta t$ should approximate the DT model, and we verify that the two produce similar equilibrium outcomes.

C Extensions

This appendix discusses several extensions of the baseline commitment model developed in Section 4. Each extension demonstrates the robustness of the core mechanism — the re-emergence of Kuhn-Tucker conditions at commitment points — and points toward applications beyond the time-to-build setting.

C.1 The Delay Differential Equation Formulation

The hybrid formulation in Section 4 restricts investment to discrete commitment points spaced τ apart. The fully general formulation allows a continuous flow of investment commitments, each with a delay of τ before installation. The capital law of motion becomes a *delay differential equation* (DDE):

$$\mathrm{d}k_t = \big[i(t-\tau) - \delta k_t\big]\mathrm{d}t. \tag{8}$$

The state of an individual agent is no longer finite-dimensional. At time t, the agent's future capital depends on the entire *pipeline* of past investment commitments $\{i(s)\}_{s\in[t-\tau,t]}$. The natural state space is therefore $(a_t, k_t, z_t, \{i(s)\}_{s\in[t-\tau,t]})$, an element of $\mathbb{R}\times\mathbb{R}_+\times\mathcal{Z}\times L^2([0,\tau])$.

The Hamilton-Jacobi-Bellman equation in this setting is defined on an infinite-dimensional state space. Following Fabbri et al. (2017), the value function satisfies a PDE in a Hilbert space:

$$\rho V(a,k,z,\phi) = \max_{c,i\geq 0}\Big\{u(c) + D_aV\cdot[rk+wz-c-i] + D_kV\cdot[\phi(0)-\delta k] + D_\phi V\cdot\dot{\phi} + \mathcal{A}_zV\Big\}, \tag{9}$$

where $\phi\in L^2([0,\tau])$ represents the pipeline, with $\phi(s)$ being the investment committed s periods ago, and $D_\phi V$ is the Fréchet derivative with respect to the pipeline function.

The key observation is that the investment choice $i\geq 0$ in equation (9) is subject to the liquidity constraint $a-i\geq \underline{a}$, which generates a Kuhn-Tucker condition at every instant — not just at discrete commitment points. The first-order condition for i is:

$$D_aV(a,k,z,\phi) = D_\phi V(a,k,z,\phi)\big|_{s=\tau} + \mu, \quad \mu\geq 0, \quad \mu(a-i-\underline{a}) = 0. \tag{10}$$

The term $D_\phi V|_{s=\tau}$ is the marginal value of adding a unit to the end of the pipeline, which will not install for τ periods. The KT multiplier μ is positive whenever the agent would like to invest more but is liquidity-constrained. This condition is the infinite-dimensional analogue of the Phase 2 KT condition in the hybrid model.

The DDE formulation confirms that the re-emergence of Kuhn-Tucker is not an artifact of the hybrid (discrete commitment) approximation. It is intrinsic to the

economics of commitment with delay.

We refer the reader to Fabbri et al. (2017) for the technical framework and to Bambi (2008) and Bambi et al. (2020) for applications of DDE methods in macroeconomics. The computational treatment of the infinite-dimensional HJB is an active area of research and is beyond the scope of this paper.

C.2 Stochastic Build Time

The baseline model treats τ as a deterministic parameter. In practice, build times are uncertain: construction projects face delays, R&D timelines are inherently random, and delivery times for capital goods vary with supply conditions.

Consider a generalization where the build time is a random variable $\tilde{\tau}$ drawn from a distribution H on $[\underline{\tau}, \bar{\tau}]$ at the time of commitment, with $\underline{\tau} > 0$. The agent learns $\tilde{\tau}$ only at the time of investment, but cannot adjust the commitment after it is made.

The commitment-point optimization becomes:

$$V(a, k, z; \tau) = \max_{i \geq 0} \mathbb{E}_{\tilde{\tau}}\big[V(a - i, k, z; \tilde{\tau})\big] \quad \text{s.t.} \quad a - i \geq \underline{a}, \tag{11}$$

where $V(a - i, k, z; \tilde{\tau})$ is the continuation value given a random installation delay.

The KT multiplier at the commitment point is:

$$\mathbb{E}_{\tilde{\tau}}\big[V_a(a - i, k, z; \tilde{\tau})\big] = \mu, \quad \mu \geq 0, \quad \mu(a - i - \underline{a}) = 0. \tag{12}$$

Stochastic build time *strengthens* the commitment friction. Uncertainty about $\tilde{\tau}$ increases the variance of the agent's future liquidity position, widening the set of states where $\mu > 0$. The precautionary under-investment result of Proposition 4.7 is therefore reinforced when build times are random.

C.3 Partial Reversibility

If the agent can partially reverse a commitment — for example, by canceling a construction project at a penalty λi, where $\lambda \in [0, 1]$ is the fraction of investment lost upon cancellation — the model interpolates between full commitment ($\lambda = 0$, no cancellation) and full flexibility ($\lambda = 1$, free cancellation restoring the investment).

With partial reversibility, the agent solves a stopping problem during the build

phase: at each instant $s \in (0, \tau)$, the agent may cancel the project and recover $(1 - \lambda)i$. The option to cancel reduces the effective commitment cost and therefore reduces the KT multiplier at the commitment point. In the limit $\lambda \to 1$ (costless cancellation), the commitment is fully reversible and the model collapses to the standard CT formulation with $\mu = 0$.

This extension confirms that the mechanism is driven by the *degree* of commitment, not just its presence. The parameter λ provides a second dimension of variation, complementing τ.

C.4 Multiple Commitment Types

In practice, agents make multiple types of commitments simultaneously: short-term equipment purchases (τ_1 small), medium-term construction (τ_2 moderate), and long-term R&D (τ_3 large). The model extends naturally to multiple investment types $j = 1, \ldots, J$ with heterogeneous build lags τ_j.

The state space expands to include a timer for each active commitment. At each commitment point for type j, a separate KT condition applies:

$$V_a = \mu_j, \quad \mu_j \geq 0, \quad \mu_j(a - i_j - \underline{a}) = 0, \quad j = 1, \ldots, J. \tag{13}$$

The interaction between commitment types generates portfolio effects: agents substitute toward short-lag investments when liquidity is tight, even if long-lag investments have higher expected returns. This composition effect is absent from both the standard CT model (all $\tau_j = 0$) and the standard DT model (all $\tau_j = \Delta t$).

C.5 Connection to the Krusell-Smith Forecasting Problem

In the standard heterogeneous-agent model, aggregate dynamics are summarized by the cross-sectional distribution Φ_t of agents over individual states. Krusell and Smith (1998) showed that a small number of moments of Φ_t (typically mean capital) suffice to forecast future prices.

With time-to-build, the aggregate state includes not only Φ_t but also the *aggregate pipeline* $Q_t = \int i(t - s)\mathrm{d}\Phi_t$ for $s \in [0, \tau]$: the distribution of committed investments that have not yet installed. The pipeline is a predictable component of future capital supply and therefore affects the forecasting problem.

In particular, current aggregate capital K_t is no longer a sufficient statistic for future prices, because the pipeline Q_t determines how much additional capital will arrive over $[t, t+\tau]$ regardless of current decisions. This suggests that models with time-to-build may require additional state variables in the Krusell-Smith forecasting rule, potentially increasing the computational burden but also enriching the aggregate dynamics with a propagation mechanism through the pipeline.

A full treatment of this extension is left to future work.

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